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Quantum computing and the implementation of precision medicine



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Precision medicine aims to tailor healthcare by integrating individual genetic, epigenetic, transcriptomic, proteomic, and clinical data, collectively referred to as multi-omic data. However, the scale and complexity of such multi-omics datasets challenge classical computing approaches. Quantum computing, which leverages superposition and entanglement (quantum-level correlations between particles), offers a fundamentally new paradigm for accelerating molecular simulations, biomarker discovery, and high-dimensional data analysis. This review explores the convergence of quantum computing and its potential to provide unmet needs in precision biomedicine research, with emphasis on applications in diagnostic modeling, multi-omic data integration and drug discovery. We highlight early proof-of-concept studies demonstrating the use of quantum machine learning for disease prediction, quantum algorithms for protein folding, and quantum generative models for novel drug design. Hybrid quantum-classical workflows are also already enabling gene network inference and prioritization of variants of uncertain significance, the latter of which is a major focus of multi-omic research worldwide. Emerging directions include digital twin simulations and real-time clinical decision support powered by quantum models. Looking ahead, the long-term vision for quantum computing in biomedicine involves in silico modeling of entire biological systems to simulate cellular responses to perturbations like drug treatments, enabling clinicians to test therapies in virtual patients before real-world application. Despite these advances, practical implementation remains limited by hardware constraints, qubit decoherence, algorithm scalability, and regulatory barriers. Nonetheless, as quantum hardware evolves and AI-aligned quantum algorithms mature, their integration holds transformative potential. Quantum computing may eventually shorten diagnostic timelines, improve therapeutic precision, and make biomedical innovation more globally accessible. We outline a roadmap for translating these technologies into next-generation precision medicine.

Precision medicine aims to tailor disease prevention and management by leveraging an individual's unique biomarker profile, including genetic, molecular, and clinical characteristics. Achieving this vision requires the efficient integration of complex, high dimensional data from multiple sources such as multi-omics datasets encompassing genomic, transcriptomic and proteomic information, alongside real-time diagnostics and advanced computational pipelines¹. Historically, treatment

strategies have evolved from genetic testing of individual genes or panels to genome-wide approaches that can simultaneously analyze all genes. This paradigm shift has given rise to the term 'genomic medicine'². Unlike traditional "one-size-fits-all" approaches, precision medicine necessitates the seamless integration of diverse patient data, from genetic variants and molecular expression profiles to clinical histories into a coherent framework³.

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However, as these datasets continue to grow in scale and complexity, conventional high-performance computing (HPC) architectures are increasingly constrained by combinatorial growth of parameters and features. This computational bottleneck has sparked interest in quantum computing as a potential solution. Recent advances in both hardware and theoretical algorithms positions quantum computing as a promising complement, or even a substitute, for classical HPC in realizing the full potential of precision medicine (Table 1a). Figure 1 presents a timeline of major technological and scientific breakthroughs in both domains, from early discoveries like the double helix structure of DNA and the development of quantum algorithms, to more recent milestones such as the approval of CRISPR therapies and the demonstration of quantum supremacy. These parallel advances suggest that quantum computing and precision medicine are on converging trajectories, poised to address the growing demands of biomedical complexity.

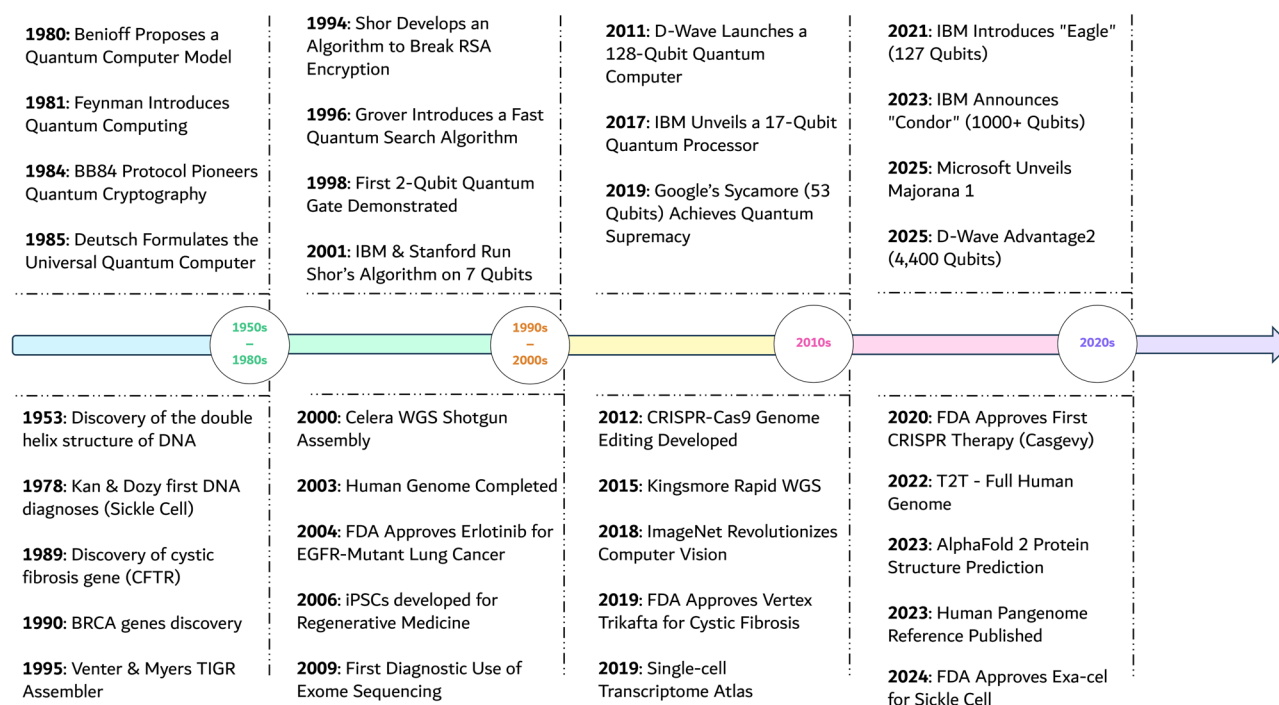
Originally rooted in physics and theoretical computer science, quantum computing was conceived to simulate quantum systems that classical computers cannot efficiently handle. Richard Feynman first proposed the use of quantum systems to simulate quantum mechanics, laying the foundation for today's quantum devices⁴. Early milestones such as boson sampling and quantum annealing targeted complex physical problems, like energy state calculations and large-scale optimization tasks^{5,6}.

Quantum devices leverage qubits, which utilize the principles of superposition and entanglement, allowing them to represent multiple states simultaneously (Box 1). This enables quantum algorithms to explore large search spaces exponentially faster than their classical counterparts in certain applications, offering a potential leap forward in addressing computationally intensive challenges⁷. The parallel evolution of quantum computing and precision medicine, shaped by foundational breakthroughs in both fields, is now converging to address the growing demands of biomedical complexity.

Table 1 | Overview of Leading Quantum Computing Platforms and Associated Software Frameworks

A. Major Quantum Computers and Their Access Levels				
Institution	Quantum Computer Name	Qubit Count	Technology / Architecture	Access Level
D-Wave	Advantage	5000+	Quantum Annealing	Limited Public Access
IBM	Condor	1121	Superconducting – Gate-based	Limited Public Access
	Heron	133	Superconducting – Gate-based	Limited Public Access
	Eagle	127	Superconducting – Gate-based	Limited Public Access
Microsoft Azure Quantum Affiliated Platforms	PASQAL	~100	Neutral Atoms (Optical Lattice) – Gate-based	Limited Public Access
	Quantinuum	20–56	Trapped Ion – Gate-based	Limited Public Access
	IonQ	29–32	Trapped Ion – Gate-based	Limited Public Access
	Rigetti	9–84	Superconducting – Gate-based	Limited Public Access
Microsoft	Majorana-1	~67–70	Topological Qubits – Gate-based	Proprietary
Harvard / MIT / QuEra	QuEra QPU	~280	Neutral Atoms – Gate-based	Academic Collaboration
University of Maryland	UMD Ion Trap QEC System	~228	Trapped Ion – QEC	Academic Collaboration
ETH Zurich / PSI	ETH Neutral Atom QEC Array	~200	Neutral Atoms – Gate-based	Research Access
Google	Willow	105	Superconducting – Gate-based	Proprietary
	Sycamore	53	Superconducting – Gate-based	Proprietary
University of Science and Technology of China (USTC)	Zuchongzhi 3.0	105	Superconducting – Gate-based	Proprietary
Intel	Tangle Lake	49	Superconducting – Gate-based	Research Access
Alibaba / Xanadu	Xanadu	24	Photonic Qubits (Continuous Variable) – Gate-based	Limited Public Access
University of Innsbruck	AQT Ion-Trap	~24	Trapped Ion – Gate-based	Research Access
B. Open-source Quantum Software Packages				
Package Name	Language (s)		Compatible Platforms	
Qiskit	Python		IBM Q	
Cirq, Quantum Echoes	Python		Google	
Ocean SDK	Python		D-Wave, Cross Platform	
QuTip	Python		Simulations (Open Quantum Systems)	
Q#	C#, Python, F#		Microsoft Azure	
PyQuil	Python		Rigetti (Forest SDK)	
Quipper	Haskell		Cross-platform	
TensorFlow Quantum (TFQ)	Python		Google, IBM, Simulations	
ProjectQ	Python		IBM, Google, Rigetti, IonQ	
cuQuantum	C++		NVIDIA GPU Accelerated QC Simulation	
PennyLane	Python, C++		IBM, Rigetti, Google, IonQ	
QMLCore	Python		Orquestra (Zapata)	
OpenQASM	C, Python		IBM Q	
Liquid	C#, NET		Microsoft Research	
Strawberry Fields	Python		Xanadu (Photonic)	

Quantum Computing



Precision Medicine

Fig. 1 | Historical milestones related to precision medicine and quantum computing. This timeline highlights major technological and scientific breakthroughs that have shaped the fields of quantum computing (top panel) and precision

medicine (bottom panel) and from the 1950s to the 2020 s. The convergence of these two domains is expected to accelerate discovery and translation research in the biomedical sciences.

Proof-of-concept studies in quantum machine learning (QML) have further demonstrated the feasibility of quantum-enhanced classifiers for biomarker discovery and patient stratification. These advancements suggest that quantum systems could unlock new frontiers in data-driven clinical decision making^{8,9}.

In summary, by leveraging the unique computational advantages of quantum systems, precision medicine stands to overcome many of its current limitations in biomedical data analysis. This integration could markedly accelerate the development of more patient-specific, efficient, and accessible healthcare solutions. Looking ahead, one can envision a future in which quantum computing not only enhances diagnostics and therapeutic design but also democratizes drug discovery, particularly in settings with limited computational infrastructure, such as developing regions or institutions with constrained access to high-performance computing resources.

Quantum computing

Quantum qubits, circuits and errors

Unlike classical bits, which are restricted to discrete values (0 s or 1 s), quantum computers utilize qubits, which can exist in a state of superposition, embodying both states simultaneously. This fundamental property, coupled with entanglement, a quantum phenomenon where the states of two or more qubits become intrinsically linked such that the state of one instantly influences the other regardless of distance, allows qubits to represent and process a multiplicity of states concurrently, effectively storing more information than a single classical bit (Box 1). Consequently, quantum algorithms can achieve exponential speedups for specific tasks. Examples include Grover's search algorithm and quantum annealing, which demonstrate efficiency in applications such as unsorted database searches and combinatorial optimization¹⁰. Theoretically, a qubit exists in a two-dimensional complex Hilbert space. When

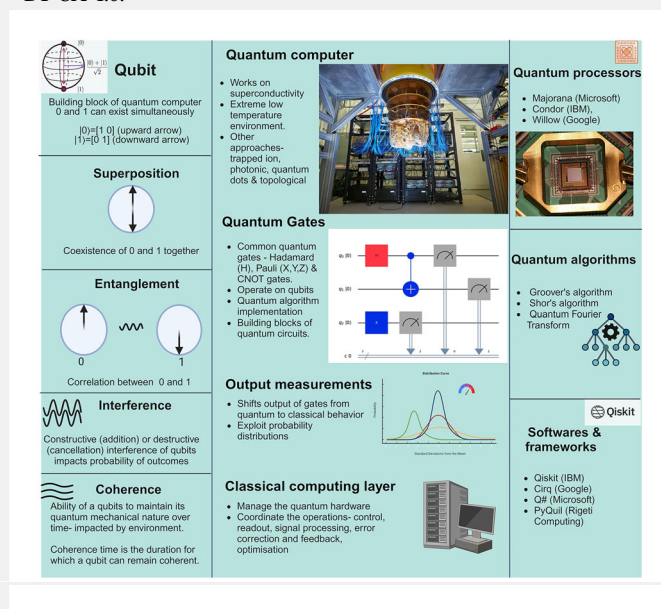
multiple qubits are entangled, the overall system scales exponentially: n qubits span a 2^n -dimensional space. This contrasts with classical systems, where capacity scales linearly. To build larger systems, qubits are often organized into modular units (*qubit modules*) that support scalability while managing coherence and error correction. While such modules do not simulate all possible quantum states, they enable architectures that have surpassed 1000 qubits in total count¹¹, signaling progress toward quantum systems that outperform classical ones for specific computational tasks. Unlike classical circuits, which process finite registers sequentially or in parallel, quantum circuits can scale exponentially, making them well-suited for computationally intensive tasks such as factorization, database searching, and molecular simulations¹².

However, quantum computing also faces challenges (also discussed below). A key limitation is decoherence, the tendency of quantum states to degrade due to environmental interactions, leading to the collapse of superposition and entanglement. Coherence refers to the system's ability to preserve quantum states over time; the longer the coherence time, the more reliably computations can be performed. Noise and decoherence may introduce significant errors that limit the scalability and accuracy of current devices. This could be particularly problematic in applications in precision medicine, where even minor errors, such as misclassifying a pathogenic variant, resulting in ineffective or delayed treatments. To mitigate these challenges, researchers are actively developing quantum-error-correction (QEC) techniques, such as topological surface codes. These methods organize qubits on a lattice and perform periodic error detection and correction, making surface codes one of the most promising strategies for achieving scalable, fault-tolerant quantum computing¹³. Additionally, techniques like Zero-Noise Extrapolation (ZNE) systematically measure quantum circuit outcomes at varying noise levels and extrapolate them to an ideal "zero-noise" case. This approach improves the accuracy of computations on near-term quantum hardware¹⁴.

Box 1 | Fundamental qubit properties, hardware, and algorithms in quantum computing

This figure provides an integrated overview of quantum computing, illustrating fundamental qubit properties, hardware components and quantum algorithms. Qubits exhibit key quantum phenomena such as superposition (simultaneous existence in states 0 and 1), entanglement (correlation between qubits), interference (modulation of probabilities), and coherence (stability of quantum states over time). These properties enable quantum computers to perform highly parallel computations. The hardware section highlights quantum processors, including Majorana-based qubits (Microsoft), superconducting qubits (IBM's Condor), and photonic qubits (Google's Willow). These systems rely on specialized environments such as extreme low temperatures, trapped ions and topological configurations, for stable quantum operations. Quantum gates, such as Hadamard, Pauli (X, Y, Z), and CNOT, execute fundamental logical operations on qubits, enabling the implementation of complex quantum algorithms. The classical computing layer manages quantum hardware, handles control operations, processes errors, and ensures feedback and optimization. The algorithms section features key quantum algorithms, including Grover's algorithm (for efficient database searches), Shor's algorithm (for prime factorization), and the Quantum Fourier Transform (for advanced signal processing). Additionally, widely used quantum software frameworks, including IBM's Qiskit, Google's Cirq, Microsoft's Q#, and Rigetti's PyQuil facilitate the development, simulation, and execution of quantum programs. Panel created in BioRender. Jamalalail, B. (2025) <https://BioRender.com/gyyr5q3>. Image attribution: reproduced from Jimmy Chen⁷⁹, American Journal of

Computational Mathematics, 2025, under the Creative Commons Attribution 4.0 International License (CC BY 4.0). The D-Wave TwoX quantum processor chip photo is by Mwjohnson0, licensed under CC BY-SA 4.0.



Topological qubits and emerging hardware

Topological qubits represent a promising approach for mitigating quantum errors by encoding and manipulating information within topological states of matter. This approach theoretically makes qubits more resilient to local disturbances and environmental noise, enhancing their stability and coherence¹⁵. Microsoft has advanced this field through the development of “Majorana-based” qubits, claiming an error rate of ~1%¹⁶. This reduced error rate is attributed to built-in hardware-level error correction, where quantum information is inherently protected against decoherence and gate errors at the fundamental hardware level, minimizing reliance on software-based correction strategies.

Beyond Microsoft's progress, emerging platforms such as Google's Willow^{17,18} highlight alternative advancements in scalable quantum hardware. Willow utilizes photonic qubits and demonstrates the potential for improved error rates as system complexity increases. As the number of qubits grows, robust control protocols and calibration strategies become crucial to maintaining hardware reliability and minimizing errors. Recent studies emphasize that while short-term advancements focus on error mitigation techniques, achieving truly fault-tolerant quantum systems remains the long-term goal for enabling clinical-grade applications. Additionally, superconducting qubits, such as those in IBM's Condor processor¹⁹, offer another pathway toward scalable quantum systems. These qubits operate at cryogenic temperatures and require precise control to reduce error rates, with advancements aimed at increasing qubit connectivity and reducing decoherence.

For quantum computing to be fully integrated into critical applications like diagnostic pipelines or drug discovery workflows, hardware must reach fault-tolerance levels that ensure consistent and error-free operations. Such advancements would enable quantum systems to handle complex computations, such as large-scale simulations in biomedical research, without the risk of catastrophic error propagation.

Quantum simulators, algorithms and packages

Quantum simulators are pivotal for modeling complex quantum systems, enabling the validation of quantum algorithms and hardware architectures. Categorized into analog (mimicking specific quantum systems) and digital (emulating systems via quantum circuits), these simulators are advancing applications in precision medicine and drug discovery. Key quantum algorithms, such as Grover's algorithm for accelerating search processes¹⁰, Shor's algorithm for efficient factorization¹⁰, and the Quantum Fourier Transform (QFT) for spectral analysis⁷, are central to advancing computational biology and chemistry. Additionally, quantum phase estimation (QPE) is instrumental in estimating molecular energy levels, crucial for drug discovery¹⁴.

Several open-source quantum software frameworks have been developed to support algorithm design and execution (Table 1b). However, a key challenge in adopting quantum computing, especially for researchers accustomed to high-level abstractions in High Performance Computing (HPC) is that quantum development still often requires low-level circuit-based programming. This represents a considerable shift from HPC's decades of mature software ecosystems. To address this, platforms such as IBM's Qiskit facilitates prototyping quantum algorithms and quantum chemistry simulations, while Google's Cirq is tailored for circuit design on its quantum hardware²⁰. Microsoft's Q# supports algorithms involving error correction and topological qubits²¹, and Rigetti's PyQuil enhances hybrid quantum-classical workflows²². For quantum machine learning, TensorFlow Quantum (TFQ) integrates quantum computing with classical machine learning frameworks²³, and PennyLane supports variational quantum algorithms essential for quantum-enhanced data analysis²⁴. These platforms are critical for accelerating algorithm development, ensuring robust quantum-classical integration, and advancing quantum applications in biomedical research and precision medicine.

Applications of quantum computing

Quantum computing for solving complex problems

The integration of big data and multi-omics research in precision medicine necessitates computational frameworks capable of deciphering high-dimensional patterns, which are often elusive due to insufficient training data or weak underlying biological signals²⁵. In such scenarios, classical machine learning (ML) models may struggle to achieve reliable accuracy, especially when key molecular or clinical features are obscured within vast datasets. Hybrid quantum-classical pipelines offer a pragmatic solution by performing large-scale data preprocessing, such as cleaning, normalization and dimensionality reduction on classical servers, while delegating complex optimization and advanced feature selection tasks to specialized quantum subroutines.

A critical benchmark for assessing the performance of quantum computers is the validation of random circuit sampling (RCS) in noisy environments, particularly their ability to surpass the capabilities of classical supercomputers. This protocol involves generating random quantum circuits and evaluating how faithfully the quantum device reproduces output distributions that are exponentially challenging for classical systems to simulate. Successful RCS demonstrations not only indicate progress toward quantum advantage but also serve as stress tests under realistic noise and decoherence conditions²⁶.

Recent advancements in quantum computing further emphasize progress toward overcoming current limitations. Notably, Google's QC Willow quantum computing platform recently demonstrated an estimated fidelity of 1.5×10^{-3} in random circuit sampling (RCS) experiments²⁷, surpassing previous benchmarks²⁶. In this context, fidelity refers to the degree of alignment between the experimentally measured quantum distribution and the theoretical ideal, with higher fidelity indicating superior noise management and more accurate gate operations. Although a fidelity of $\sim 10^{-3}$ still falls within the noisy intermediate-scale quantum (NISQ) regime, such milestones underscore significant advancements in qubit quality, error correction strategies, and hardware architectures. These improvements are critical for moving closer to fault-tolerant quantum computing, ultimately enhancing the reliability of quantum systems in tackling complex problems, including those in precision medicine.

Quantum computing in precision medicine

Quantum computing is poised to transform precision medicine by addressing computational barriers that hinder individualized healthcare. This section highlights key applications where quantum algorithms and devices are being explored, with a focus on diagnostics, drug development, protein engineering, and systems biology. Quantum computing offers novel solutions for tackling the complexities of biomedical data, particularly through the principles of superposition and interference. These principles enable quantum algorithms to efficiently traverse vast search spaces, an advantage especially relevant when classical machine learning (ML) approaches struggle due to sparse training data or combinatorial complexity. For instance, quantum algorithms can accelerate biomarker discovery and disease subtyping by effectively pruning non-optimal solutions and narrowing the search space²⁸.

Quantum annealing techniques have demonstrated promise in optimizing complex biological problems, such as protein folding and biomarker selection. By mapping these optimization tasks onto quantum Hamiltonians, improved search efficiency has been observed in pilot studies, although hardware limitations currently restrict large-scale clinical^{29,30}. Additionally, the use of entangled qubits can facilitate compact representation of complex feature spaces, enhancing tasks such as differentiating between malignant and benign lesions in imaging, or filtering relevant signals from multi-omic datasets. Numerous proof-of-concept studies have successfully applied quantum computing to tasks such as gene regulatory network inference, sequence alignment, and disease prediction (Table 2). These advancements highlight the potential of quantum computing to overcome key bottlenecks in biomedical data processing and analysis,

driving the development of more precise, efficient, and scalable solutions in precision medicine.

Quantum computing for diagnostics and prognosis. Precision medicine relies heavily on early diagnosis and accurate prognosis to guide informed treatment decisions. Quantum computing offers transformative potential in this domain by addressing limitations in classical algorithms, especially in processing high-dimensional clinical data and recognizing subtle disease signatures.

Quantum Machine Learning (QML) models are proving particularly effective in enhancing diagnostic accuracy. For example, a hybrid quantum-classical neural network applied to mammogram images improved tumor classification accuracy from approximately 77% to 84%, demonstrating the feasibility of real-hardware quantum diagnostics³¹. Nadine et al.³² developed a nine-qubit Quantum Convolutional Neural Network (QCNN) that matched classical CNN performance while offering novel feature extraction capabilities via quantum gates. A separate QCNN model for lung cancer detection using CT scans achieved 92%, outperforming the best classical baseline at 89% integrating variational quantum circuits with pre-trained classical models³³. Likewise, a Hybrid Quantum CNN (HQCNN) tested on over 7000 brain MRI samples showed marginal but consistent improvements over traditional CNNs on both Kaggle and REMBRANDT datasets (e.g., 97% vs 96% precision)³⁴, further showcasing QML's advantages in image-based diagnostics.

Quantum computing also shows promise in advancing prognosis models. An Improved Quantum CNN (IQCNN) designed for heart disease prediction across three datasets consistently outperformed classical baselines of 80%, achieving accuracy rates as high as 93%³⁵. A Quantum Support Vector Machine (QSVM) for prostate cancer detection demonstrated 92% accuracy comparable to its classical counterpart but achieved 100% sensitivity (7% increase)³⁶. Additionally, Quantum-Enhanced Machine Learning (QuEML) models applied to the Kaggle heart disease dataset slightly outperformed traditional models in both accuracy (96% vs 95%) and training speed (192.5 μ s faster), underscoring its computational efficiency and practical utility³⁷.

Collectively, these advancements suggest that quantum computing can significantly elevate diagnostic accuracy and early disease detection. The ability to process complex, high-dimensional datasets more efficiently will be instrumental in guiding individualized treatment strategies and enhancing prognostic predictions. As quantum algorithms and hardware continue to mature, their integration into diagnostic workflows holds promise for revolutionizing precision medicine by not only enhancing both accuracy and speed, but also by potentially reducing the growing energy demands associated with traditional high-performance computing infrastructures¹⁸.

Quantum computing for multi-omics and systems biology. Precision medicine increasingly relies on the integration of diverse omics layers, including genomics, epigenomics, transcriptomics, proteomics, and metabolomics to reveal holistic insights into disease mechanisms and identify predictive biomarkers. However, integrating and analyzing such high-dimensional, multi-omics data remains a formidable computational challenge. Identifying disease-associated molecular signatures often involves evaluating an astronomically large number of feature combinations, far beyond the reach of classical algorithms. Quantum computing offers a promising solution by enabling efficient exploration of these vast combinatorial spaces. Quantum algorithms, particularly those designed for machine learning and optimization, can evaluate numerous biomarker combinations in parallel. They also hold promise for quantum-enhanced clustering of patients across thousands of molecular features, potentially revealing hidden subpopulations or outliers that classical methods may overlook.

Significant progress has been made in systems biology applications, particularly in gene regulatory network (GRN) inference, which is central to understanding gene-gene interactions and regulatory mechanisms. A 2023 study introduced a hybrid quantum-classical algorithm called qscGRN

Table 2 | Key Applications of Quantum Computing in Precision Medicine and Biomedical Research

Application	Description	Access Link
QuantumGRN ³⁸	Quantum circuit framework to infer gene regulatory networks from single-cell transcriptomic data	https://github.com/caiilab-tamu/QuantumGRN
scRNA-seq-Clustering-QA ⁵¹	Uses quantum annealing to optimize clustering cost functions in single-cell RNA-seq data (via D-Wave)	https://github.com/michal7kw/scRNA-seq-Clustering-QA
QGAN-HG ⁵⁴	Hybrid quantum GAN for efficient small molecule generation and drug design, validated on IBM quantum hardware	https://github.com/jundeli/quantum-gan
QuASer ⁵⁵	Quantum-based DNA sequence reconstruction (de novo assembly)	https://github.com/QE-Lab/QuASer
QuEST ⁵⁶	Toolkit for exact quantum pattern matching in biological sequences	https://github.com/profkapilsoni/EnQPBEA-and-EnQBCEA-Algorithms
D-Wave-VRP ⁵⁷	Hybrid quantum-classical approach for DNA sequence assembly using D-Wave	https://github.com/dwave-examples/D-Wave-VRP
Qiskit ⁵⁸	Implements a quantum protein folding algorithm optimized for resource efficiency	https://github.com/Qiskit
MATLAB Plugin ⁵⁹	Quantum cuckoo search algorithm for multiple sequence alignment	https://www.mathworks.com/products/quantum-computing.html
lambeq ⁶⁰	Applies quantum natural language processing to protein sequence analysis	https://github.com/CQCL/lambeq
Fujitsu Digital Annealer ⁶¹	Quantum-inspired graph-cut approach for phylogenetic tree construction	https://portal.aispf.global.fujitsu.com/apidoc/da/jp/api-ref/da-qubo-en.html
TFQ-QCNN ⁶²	Uses Quantum CNNs for ischemic heart disease prediction, improving early diagnosis	https://www.tensorflow.org/quantum/

that successfully inferred GRNs from single-cell gene expression data³⁸. The quantum-enhanced model captured higher-order dependencies more accurately than classical approaches and revealed novel gene interactions, several of which were later validated by biological experts. Such insights have profound implications for precision medicine, where identifying key regulatory genes or pathways could lead to more targeted and effective interventions.

Further advancements include quantum-assisted marker selection in single-cell multi-omics data. A recent study introduced Bisection Grover's Search (BGS) for optimizing biomarker selection in CITE-seq datasets that simultaneously measure RNA and protein expression at the single-cell level³⁹. By integrating Grover's algorithm with binary partitioning, BGS achieved faster and more accurate identification of antibody-derived tags (ADTs) than classical techniques. Its successful implementation on both IBM quantum hardware and simulators underscores the near-term feasibility of quantum-enhanced omics analysis.

These advances highlight the potential of quantum computing to accelerate multi-omics integration and systems-level biological inference. As quantum hardware continues to improve, these tools may become integral to deciphering complex disease phenotypes and advancing data-driven precision medicine.

Quantum computing for drug discovery. Drug discovery is a complex, data-intensive process, often limited by the computational constraints of classical methods. Quantum computing presents a transformative opportunity by enabling realistic simulations of molecular dynamics and drug-target interactions with high precision. This capability is particularly vital in precision medicine, where individualized drug responses hinge on diverse genetic, epigenetic, and environmental factors. Classical algorithms encounter scaling limitations in molecular simulations as system size and complexity increase. Quantum computers, by contrast, natively represent molecular quantum states, allowing them to efficiently manage these complexities. Variational quantum algorithms can accurately estimate binding energies and reaction dynamics, providing early predictions of drug-target interactions before laboratory synthesis.

Recent studies highlight the potential of integrating machine learning (ML) with quantum computing to enhance drug discovery. ML models, such as recurrent neural networks (RNNs), generative adversarial networks (GANs), and variational autoencoders (VAEs), have shown promise in generating molecular structures optimized for binding affinity^{40–42}.

However, classical high-performance computing (HPC) struggles to scale these approaches to massive molecular libraries. Quantum computing can overcome these limitations, offering faster optimization and pattern recognition to accelerate the identification of potential drug candidates.

A landmark study⁴³ demonstrated the use of quantum-enhanced methods to identify 15 inhibitory molecules with high potential for KRAS binding, a critical driver in multiple cancers. The analysis, which would have required significantly greater computational time and energy using traditional high-performance computing (HPC) approaches, leveraged a hybrid quantum-classical workflow. Classical preprocessing was performed on GPU nodes, while quantum computations were executed on a 16-qubit Guadalupe quantum system. Notably, the study reported a linear correlation between the number of qubits and prediction success rate, underscoring the scalability of their approach. The Quantum Circuit Born Machine (QCBM) model was employed for error correction, and variational quantum circuits were combined with classical LSTM models for optimization. The Optuna platform facilitated hyperparameter tuning, ultimately enabling the identification of high-affinity lead candidates with improved accuracy and computational efficiency compared to classical pipelines, demonstrating why this study marks a significant advancement in quantum drug discovery.

Addressing the challenge of representing structurally diverse compound libraries, another study⁴⁴ utilized a quantum support vector machine (QSVM) using Qiskit to compress inhibitor descriptors for diseases like SARS-CoV-2 and Mycobacterium tuberculosis. Deployed on an IBM quantum computer with 53 qubits, the QSVM achieved accuracy comparable to classical PCA methods while enhancing computational efficiency. Further demonstrating the potential of quantum computing, a hybrid approach integrating quantum and classical systems achieved significant reductions in computation time. This method scaled AI-driven discovery across large molecular datasets, illustrating the practical advantages of quantum systems in accelerating drug discovery workflows. These advancements underscore the potential of quantum computing to overcome key bottlenecks in drug discovery, facilitating faster and more accurate identification of viable therapeutics and advancing the prospects of precision medicine.

Quantum computing for predicting molecular structure and interactions. Accurate prediction of protein structure and molecular interactions is foundational to rational drug design, vaccine development, and

understanding biological function. While classical deep learning approaches like AlphaFold2 have made significant strides⁴⁵, they often struggle with conformational flexibility, large protein assemblies, and dynamic folding pathways. These challenges are exacerbated by memory limitations on classical computers, particularly when modeling large or complex proteins⁴⁶. Quantum computing offers complementary capabilities by efficiently navigating the vast combinatorial space of protein folding and molecular interactions, reframing them as optimization problems solvable via quantum parallelism.

Early efforts employed quantum annealing to study protein folding using simplified lattice frameworks such as the hydrophobic-polar (HP) model. Here, amino acids are classified as hydrophobic or polar, and the folding goal is to maximize hydrophobic contacts. A D-Wave quantum annealer successfully identified ground-state configurations for small proteins with high accuracy, validating the feasibility of quantum optimization in biophysical contexts^{30,45}. Building on this, Grover's search algorithm has been applied to a hydrophobic-hydrophilic model, achieving quadratic speedup over classical techniques when implemented on IBM Quantum's QASM simulator⁴⁷. These approaches accelerate the search for low-energy protein conformations, which are key to identifying viable therapeutic targets.

Recent breakthroughs show that quantum hardware can handle more biologically relevant systems. A recent study used quantum computation to predict the structure of a catalytic loop in the Zika virus NS3 helicase, demonstrating real-world utility in drug and vaccine development⁴⁵. In parallel, hybrid quantum-machine learning frameworks are advancing molecular interaction predictions. One study combined quantum mechanics with LASSO-SVM to predict enzymatic stereoselectivity with 99.6% explanatory power, providing critical insights into enzyme-substrate binding specificity, an essential step in lead optimization⁴⁸.

As quantum hardware evolves toward fault-tolerance and higher qubit counts, fully quantum or hybrid workflows may transform the structure-based drug design pipeline. These technologies could accelerate virtual screening, optimize drug-target interactions, and refine *in silico* toxicity predictions, ultimately enabling faster, more precise, and cost-effective development of next-generation therapeutics.

Future application of quantum computing in precision medicine

As quantum computing technologies mature, their applications in precision medicine are poised to transform how diagnostics, treatment, and data interpretation are approached. Quantum computing offers a unique advantage in analyzing complex diseases, such as cancer, where high-dimensional data like imaging, cell-free circulating tumor DNA (ctDNA) profiles, and omics datasets pose significant computational challenges. For instance, ctDNA interpretation is complicated by tumor heterogeneity and diverse mutation or methylation signatures. Somatic variant signatures are more prominent in lung cancer⁴⁹, while methylation signatures dominate in breast cancer⁵⁰. Analyzing such biomarkers requires massive sequencing depth and large-scale datasets. Quantum computing, through its inherent parallelism, can accelerate this analysis, enabling more accurate predictions and faster clinical insights.

For example, in ctDNA clustering tasks, D-Wave quantum annealing models not only converged faster but also found globally optimal clusters with improved Silhouette and Rand scores compared to classical k-means approaches⁵¹. Such gains highlight the potential for quantum methods to reduce energy consumption and runtime in data-intensive clinical workflows. Classical pipelines, particularly those relying on GPU-based HPC infrastructure, can require dozens of GPUs over several days, significantly increasing energy and cooling demands. In contrast, quantum or hybrid systems can deliver similar or better performance at a fraction of the energy cost, addressing major infrastructure bottlenecks for institutions where utility and capital expenditure are key constraints.

In clinical decision-making, quantum computing could drastically improve real-time multi-omic integration. Its speed and efficiency would

enable rapid interpretation of vast genomic, transcriptomic, and epigenomic data, reducing diagnostic timelines and enhancing precision therapies. One emerging concept is the creation of digital twins, virtual patient models that simulate disease progression or treatment response using quantum-powered models of molecular dynamics and protein interactions. These could optimize therapeutic regimens with unprecedented speed and accuracy.

Another area where quantum computing could prove transformative is in the interpretation of variants of uncertain significance (VUS). VUS represents up to 40–50% of variants detected in rare disease and cancer studies⁵². Traditionally, they require time-consuming experimental validation. Quantum computing, especially quantum machine learning (QML), can rapidly correlate VUS with large-scale multi-omics profiles. Quantum algorithms, such as quantum-enhanced support vector machines or decision trees can process high-dimensional feature spaces more efficiently than classical models, clustering or scoring VUS based on molecular associations often missed by conventional methods. This could help prioritize VUS with higher pathogenic potential, significantly accelerating variant classification and downstream therapeutic development.

In parallel, initiatives like the EMBL-EU funded projects, quantum computing is being explored for large-scale pangenome analysis, an area critical for global precision medicine and biodiversity genomics. The goal is to harness quantum algorithms to process enormous genomic diversity datasets, offering faster and more precise insights into evolutionary biology, disease mechanisms, and population-level health data.

In drug discovery, quantum computing offers a transformative leap. By simulating quantum chemistry more precisely than classical methods, quantum platforms can screen millions of compounds against multiple disease targets, improving the fidelity of lead identification and reducing the high failure rates in clinical trials. Importantly, such advances could also make drug development more accessible and affordable, especially when paired with open-source platforms and global collaborations. In resource-limited settings, quantum systems have the potential to support equitable health outcomes by accelerating the development of diagnostics and therapeutics for neglected diseases.

In the long term, quantum computing might enable *in silico* modeling of whole biological systems at a fidelity not possible before. Quantum-based simulations could potentially model the concurrent interactions of thousands of biochemical reactions in a cell or predict how an entire network responds to a perturbation (like a new drug). While such applications are still speculative, the ongoing progress in quantum algorithms and hardware makes it conceivable that future quantum computers could simulate aspects of cell biology or human physiology as integrated systems. Success in this arena would be transformative: clinicians could simulate a specific patient's cellular response to a drug or genetic edit before applying it, essentially enabling virtual simulations of patient-specific clinical trials. Though significant breakthroughs are still required to realize this vision, the initial proofs-of-concept in multi-omics data analysis and network inference demonstrate the direction of the field. By leveraging quantum computing for systems-level biomedical problems, precision medicine could move from a reactive discipline (treating conditions after they arise) to a proactive and predictive one, where complex models inform early interventions tailored to an individual's unique biology.

Finally, the convergence of quantum computing and AI presents transformative opportunities. Quantum-aware AI models designed to exploit quantum architectures may enable near real-time translation of multi-omic data into clinical insights. This co-designed approach could unlock novel disease mechanisms, improve diagnostic specificity, and facilitate highly individualized treatments, thereby advancing new era of inclusive and globally equitable precision medicine.

Challenges and limitations

Despite recent advancements, several challenges must be overcome to fully harness the potential of quantum computing in precision medicine.

Table 3 | Classical Challenges in Biomedical Data Analysis and the Potential of Quantum Computing to Overcome Them

Discipline	Major Computational Challenges	Current Solutions	Potential Quantum Breakthroughs
Diagnostics and Prognosis	Noisy signals in diagnostics (MRI, CT, EEG) Capturing subtle disease progression	CNNs, radiomics, time-series ML models ⁶³ <i>Limitation:</i> Poor generalization and low interpretability	Quantum CNNs for enhanced image analysis ³³ Quantum-enhanced prognosis models (QuEML, e.g., for heart disease) Faster pattern recognition and model training
Genomics	Huge data volume from multi-omics (WGS, scRNA-seq, spatial transcriptomics, etc.) ⁶⁴ Incomplete genotype - phenotype mapping due to epistasis and environmental effects ⁶⁵ Interpretation of variants of uncertain significance	Cloud-scale pipelines (e.g., GATK, STAR) for alignment and variant calling ⁶⁶ AI models for polygenic risk and variant mining ⁶⁷ <i>Limitations:</i> Compute bottlenecks, noise and variability	Quantum search and pattern matching in genomic data ⁵⁶ Detect complex, non-linear variant-trait links ⁶⁸ Quantum subspace expansion to model epistasis ⁶⁹
Multi-omics Feature Selection	Very high dimensionality (10^4 - 10^6 features) Heterogeneous, noisy datasets Lack of consensus biomarkers	Feature selection via LASSO, RFs, XGBoost ⁷⁰ PCA, t-SNE for dimensionality reduction ⁷¹ <i>Limitation:</i> Misses complex feature interactions	Quantum SVMs/ decision trees for non-linear feature selection ⁷² Bisection Grover's Search (BGS) for optimal marker discovery in CITE-seq data ³⁹ Faster convergence to robust biomarker sets across modalities
Multi-omics Network Modeling	Modeling dynamic, non-linear cellular interactions Time-series inference of causal relationships	Bayesian networks, GRNs using mutual information ⁷³ <i>Limitation:</i> Computation-heavy for large datasets	Quantum GRN inference (qscGRN) Identifies high-order regulatory relationships missed by classical methods
Drug Discovery	Vast search space ($\sim 10^{60}$ compounds) 90%+ clinical trial failure rates ⁷⁴ Inaccurate modeling at quantum scale	ML for QSAR, docking, MD simulations ⁷⁵ High-throughput screening and lab validation ⁷⁶ <i>Limitations:</i> Approximation-reliant and slow hit-to-lead transitions ^{68,74}	Quantum solvers (VQE, QAOA) for molecular energy prediction ^{77,78} Quantum GANs (QGAN-HG) for de novo molecule generation ⁷⁸ More accurate affinity predictions for drug-target binding ⁷⁸

Hardware constraints remain a significant limitation, as current quantum devices can only support tens to hundreds of qubits, which is insufficient for handling large-scale clinical datasets. Additionally, qubits are highly sensitive to external noise, leading to decoherence and errors that can compromise complex calculations. While error mitigation strategies offer partial solutions, achieving full fault-tolerant quantum computing is still a long-term goal.

One notable concern relates to the inconsistency in claims about topological quantum computing. For instance, skepticism. The evidence, based on the topological gap protocol (TGP), relies on indirect readouts rather than direct detection of Majorana quasiparticles. Critics argue that inconsistent definitions and variable experimental parameters undermine the reliability of the results. A recent critique⁵³ highlights that Microsoft's data may not conclusively demonstrate topological behavior, casting doubt on the physical realization of such qubits. While Microsoft asserts progress, the lack of direct evidence underscores the broader challenge of validating exotic quantum states through reproducible protocols.

Another major obstacle is the scalability of algorithms, as many promising quantum algorithms assume ideal, low-error hardware. The gap between theoretical models and real-world noisy intermediate-scale quantum (NISQ) devices necessitates robust adaptations, often in hybrid quantum-classical workflows. Data integration and standardization pose further difficulties since precision medicine relies on diverse data types, including genomic variants, transcriptomic profiles, proteomic measurements, imaging, and clinical records. Encoding these heterogeneous data into qubit registers efficiently remains a challenge, and standardized pipelines for quantum data processing are still under development. These multifaceted technical and biological barriers are further summarized in Table 3, which highlights current computational limitations across key domains like genomics and drug discovery, along with how quantum computing may offer potential breakthroughs.

Additionally, validation, reproducibility, and regulatory hurdles must be addressed to ensure the clinical applicability of quantum solutions. Quantum algorithms need to be validated against gold-standard clinical benchmarks, but differences in hardware architectures and qubit fidelities complicate reproducibility. Furthermore, regulatory bodies require rigorous clinical validation of quantum-enhanced diagnostics and treatments before

approving them for routine patient care. Finally, privacy and security concerns arise with the use of quantum cloud services, as patient data is highly sensitive. Ensuring compliance with frameworks like HIPAA or GDPR requires specialized encryption and security protocols to safeguard patient information. Addressing these challenges will be crucial in making quantum computing a viable tool for precision medicine.

Conclusion

Quantum computing holds significant potential to transform precision medicine research by tackling the inherent complexity and scale of biomedical data. From multi-omics integration to liquid biopsy interpretation and targeted drug discovery, quantum algorithms can offer computational speedups or enhanced accuracy that were previously unattainable with classical methods alone. While current NISQ hardware and quantum error rates limit the scope of near-term applications, ongoing innovations in error correction, algorithmic design, and hybrid quantum-classical architectures are steadily closing this gap. Ultimately, realizing the full promise of quantum computing in healthcare will necessitate multidisciplinary collaboration among quantum physicists, biologists, computer scientists, clinicians, and regulatory bodies. By fostering robust partnerships and leveraging dedicated investment in quantum research, we can develop solutions that propel medical diagnostics, treatment planning, and drug development into a new era. As quantum devices mature toward fault tolerance, their integration into clinical workflows could help actualize the overarching vision of precision medicine: delivering the right treatment to the right patient at the right time, backed by unparalleled computational power.

Data availability

No datasets were generated or analysed during the current study.

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Author contributions

N.N. and M.U. conceived the project and structured the initial draft of the manuscript. N.N., M.A.H., K.G.R., B.J., A.M., S.W.S., A.A., and M.J.U. contributed to review of publications and to writing. N.N., M.A.H., K.G.R., B.J., and M.U. structured Figures and Tables.

Competing interests

S.W.S. is on the Scientific Advisory Committee of Population Bio and intellectual property from aspects of his research held at the Hospital for Sick Children are licensed to Athena Diagnostics and Population Bio. These relationships did not influence data interpretation or presentation during this study, but are still being disclosed for potential future considerations.

Additional information

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